Chapter 6 Review

Campbell Biology, AP Edition, 12e

Chapter 6 Review

Summary of Key Concepts

In general, eukaryotic cells have a smaller surface-to-volume ratio than prokaryotic cells, but they contain internal membrane-bound organelles. How do these internal membranes affect the overall functional efficiency of a cell and offset the low surface-to-volume ratio? **(BIG IDEAS 2 & 4)**

Concept 6.1: Biologists use microscopes and biochemistry to study cells

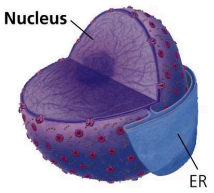
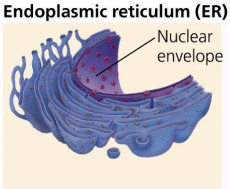
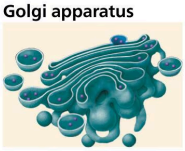
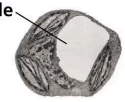
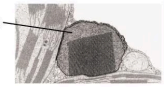
- Improvements in microscopy that affect the parameters of magnification, resolution, and contrast have catalyzed progress in the study of cell structure. **Light microscopy** (LM) and ****electron microscopy** (EM), as well as other types, remain important tools.
- Cell biologists can obtain pellets enriched in particular cellular components by centrifuging disrupted cells at sequential speeds, a process known as **cell fractionation**.

How do microscopy and biochemistry complement each other to reveal cell structure and function?

Concept 6.2: Eukaryotic cells have internal membranes that compartmentalize their functions

- All cells are bounded by a **plasma membrane**.
- **Prokaryotic cells** lack nuclei and other membrane-enclosed **organelles**, while **eukaryotic cells** have internal membranes that compartmentalize cellular functions.
- The surface-to-volume ratio is an important parameter affecting cell size and shape.
- Plant and animal cells have most of the same organelles: a nucleus, endoplasmic reticulum, Golgi apparatus, and mitochondria. Chloroplasts are present only in cells of photosynthetic eukaryotes.

Explain how the compartmental organization of a eukaryotic cell contributes to its biochemical functioning.

	Cell Component	Structure	Function
CONCEPT 6.3 **The eukaryotic cell's genetic instructions are housed in the nucleus and carried out by the ribosomes** *? Describe the relationship between the nucleus and ribosomes.*	**Nucleus**  	Surrounded by nuclear envelope (double membrane) perforated by nuclear pores; nuclear envelope is continuous with endoplasmic reticulum (ER)	Houses chromosomes, which are made of chromatin (DNA and proteins); contains nucleoli, where ribosomal subunits are made; pores regulate entry and exit of materials
	Ribosome 	Two subunits made of ribosomal RNAs and proteins; can be free in cytosol or bound to ER	Protein synthesis
CONCEPT 6.4 **The endomembrane system regulates protein traffic and performs metabolic functions** *? Describe the key role played by transport vesicles in the endomembrane system.*	**Endoplasmic reticulum (ER)**  	Extensive network of membrane-bounded tubules and sacs; ER membrane separates lumen from cytosol and is continuous with nuclear envelope	**Smooth ER:** synthesis of lipids, metabolism of carbohydrates, storage of calcium ions, detoxification of drugs and poisons **Rough ER:** aids in synthesis of secretory and other proteins on bound ribosomes; adds carbohydrates to proteins to make glycoproteins; produces new membrane
	Golgi apparatus 	Stacks of flattened membranous sacs; has polarity (*cis* and *trans* faces)	Modification of proteins, carbohydrates on proteins, and phospholipids; synthesis of many polysaccharides; sorting of Golgi products, which are then released in vesicles
	Lysosome 	Membranous sac of hydrolytic enzymes (in animal cells)	Breakdown of ingested substances, cell macromolecules, and damaged organelles for recycling
	Vacuole 	Large membrane-bounded vesicle	Digestion, storage, waste disposal, water balance, cell growth, and protection
CONCEPT 6.5 **Mitochondria and chloroplasts change energy from one form to another** *? What does the endosymbiont theory propose as the origin for mitochondria and chloroplasts? Explain.*	**Mitochondrion** 	Bounded by double membrane; inner membrane has infoldings	Cellular respiration
	Chloroplast 	Typically two membranes around fluid stroma, which contains thylakoids stacked into grana	Photosynthesis (chloroplasts are present in cells of photosynthetic eukaryotes, including plants)
	Peroxisome 	Specialized metabolic compartment bounded by a single membrane	Contains enzymes that transfer H atoms from substrates to oxygen, producing H_2O_2 (hydrogen peroxide), which is converted to H_2O

Concept	Cell Component	Structure	Function
Concept 6.3. The eukaryotic cell's genetic instructions are housed in the nucleus and carried out by the ribosomes (pages 102 to 104). Describe the relationship between the nucleus and ribosomes.	Nucleus and E R	Surrounded by nuclear envelope (double membrane) perforated by nuclear pores; nuclear envelope is continuous with endoplasmic reticulum (E R)	Houses chromosomes, which are made of chromatin (D N A and proteins); contains nucleoli, where ribosomal subunits are made; pores regulate entry and exit of materials
Concept 6.3	Ribosome	Two subunits made of ribosomal RNAs and proteins; can be free in cytosol or bound to E R	Protein synthesis

Concept 6.4. The endomembrane system regulates protein traffic and performs metabolic functions (pages 104 to 109). Describe the key role played by transport vesicles in the endomembrane system.	**Endoplasmic reticulum, E R, and nuclear envelope**	**Extensive network of membrane-bounded tubules and sacs; ER membrane separates lumen from cytosol and is continuous with nuclear envelope**	**Smooth E R: synthesis of lipids, metabolism of carbohydrates, storage of calcium ions, detoxification of drugs and poisons Rough E R: aids in synthesis of secretory and other proteins on bound ribosomes; adds carbohydrates to proteins to make glycoproteins; produces new membrane**
Concept 6.4.	Golgi apparatus	Stacks of flattened membranous sacs; has polarity (cis and trans faces)	Modification of proteins, carbohydrates on proteins, and phospholipids; synthesis of many polysaccharides; sorting of Golgi products, which are then released in vesicles
Concept 6.4.	Lysosome	Membranous sac of hydrolytic enzymes (in animal cells)	Breakdown of ingested substances, cell macromolecules, and damaged organelles for recycling
Concept 6.4.	Vacuole	Large membrane-bounded vesicle	Digestion, storage, waste disposal, water balance, cell growth, and protection

Concept 6.5. Mitochondria and chloroplasts change energy from one form to another (pages 109 to 112). What does the endosymbiont theory propose as the origin for mitochondria and chloroplasts? Explain.	**Mitochondria**	**Bounded by double membrane; inner membrane has infoldings**	**Cellular respiration**
Concept 6.5	Chloroplast	Typically two membranes around fluid stroma, which contains thylakoids stacked into grana	Photosynthesis (chloroplasts are present in cells of photosynthetic eukaryotes, including plants)
Concept 6.5	Peroxisome	Specialized metabolic compartment bounded by a single membrane	Contains enzymes that transfer H atoms from substrates to oxygen, producing H_2O_2 (hydrogen peroxide), which is converted to H_2O

Concept 6.6: The cytoskeleton is a network of fibers that organizes structures and activities in the cell

- The **cytoskeleton** functions in structural support for the cell and in motility and signal transmission.
- ****Microtubules** shape the cell, guide organelle movement, and separate chromosomes in dividing cells. **Cilia** and **flagella** are motile appendages containing microtubules. Primary cilia also play sensory and signaling roles. **Microfilaments** are thin rods that function in muscle contraction, amoeboid movement, **cytoplasmic streaming**, and support of microvilli. **Intermediate filaments** support cell shape and fix organelles in place.

Describe the role of motor proteins inside the eukaryotic cell and in whole-cell movement.

Concept 6.7: Extracellular components and connections between cells help coordinate cellular activities

- Plant **cell walls** are made of cellulose fibers embedded in other polysaccharides and proteins.
- Animal cells secrete glycoproteins and **proteoglycans** that form the **extracellular matrix (ECM)**, which functions in support, adhesion, movement, and regulation.

- Cell junctions connect neighboring cells. Plants have **plasmodesmata** that pass through adjoining cell walls. Animal cells have ****tight junctions**, **desmosomes**, and ****gap junctions**.

Compare the structure and functions of a plant cell wall and the extracellular matrix of an animal cell.

Concept 6.8: A cell is greater than the sum of its parts

Many components work together in a functioning cell.

When a cell ingests a bacterium, what role does the nucleus play?